APPSC GROUP 1 / GROUP 2 — GEOGRAPHY

Volume 4: Social, Economic and Environmental Geography; World Climate Regions

World climate regions · Population and settlements · Agriculture · Minerals and energy · Industries · Transport, communication and trade · Ecology and biodiversity · Pollution, climate change and environmental laws

Where this volume fits: Volumes 1–3 covered General Geography, Physical Geography of India and Andhra Pradesh. This volume finishes the notification's Part D: **Social Geography, Economic Geography** (World, India, State), **World climate regions**, and the **environment** items of Mains Paper II (Item 15) and Paper III. Read after Volumes 1–3.
How to read: one chapter a day. Blue box first (the idea), then tables, then orange (Andhra Pradesh link), red (exam traps), green (2-minute revision). Keep an atlas open for maps. Then attempt Geography Mock Test 4.
Data note: population figures follow Census 2011 (the latest completed census). Production rankings change yearly; learn the stable facts (regions, types, first/largest) rather than yearly tonnes.

CHAPTER 1 — WORLD CLIMATE REGIONS AND NATURAL VEGETATION

Why each part of the world has its own climate and its own plants

IN SIMPLE WORDS (what / how / why)

Climate = the average weather of a place over 30+ years. It depends on six controls: **latitude** (sun's angle), **altitude** (temperature falls about 6.5°C per 1000 m), **distance from sea** (sea keeps coasts mild; interiors have extremes), **ocean currents**, **prevailing winds** and **relief** (mountains cause rain on one side, desert on the other). Scientists group places with similar temperature and rain into **climate regions**; plants adapt to each, so every climate has its **natural vegetation**. The **Köppen** scheme (letters A, B, C, D, E and H) is the one exams use.

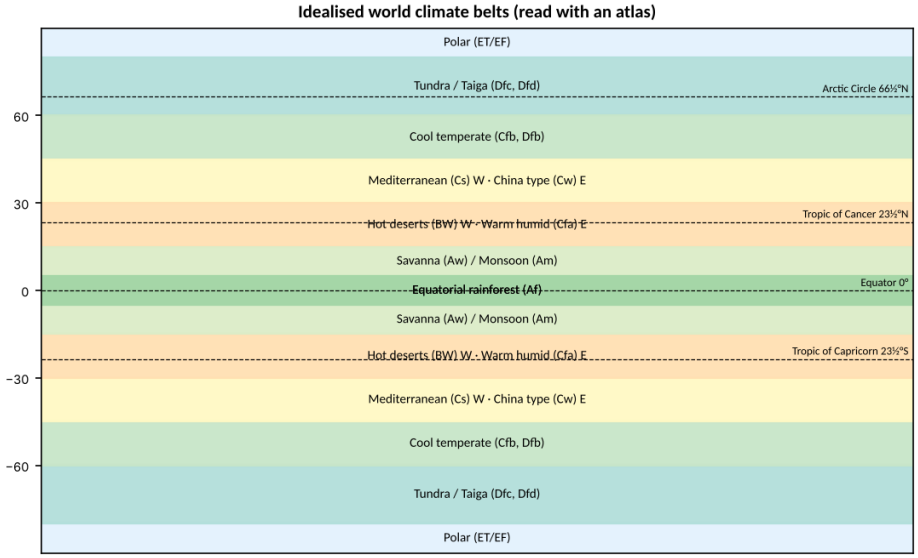


Fig 1.1 Idealised climate belts (west coasts and east coasts of continents differ)

Köppen group · Type	Where (examples)	Climate and vegetation
Af Equatorial rainforest	Amazon basin, Congo basin, Indonesia, Malaysia	Hot all year (25–27°C), rain daily (over 200 cm); evergreen broad-leaved, 3 layers, mahogany, ebony; selvas
Aw Savanna (tropical wet-and-dry)	Africa (Sudan, Kenya), Llanos (Venezuela), Campos (Brazil), N Australia	Dry winter, wet summer; tall grass with scattered trees; big grazers and predators (lion, zebra, giraffe)
Am Tropical monsoon	India, Bangladesh, Myanmar, Vietnam, SE Asia	Seasonal wind reversal; heavy summer rain; teak, sal (monsoon deciduous forests)
BWh / BSh Hot desert and semi-arid	Sahara, Arabia, Thar, Kalahari, Australian, **Atacama (driest)**	On west sides near 15–30° (trade winds blow offshore, cold currents); cactus, thorny scrub, date palm; huge day–night range
Cs Mediterranean	Around Mediterranean Sea, California, central Chile, Cape Town, SW and S Australia	**Dry summer, rainy winter** (westerlies in winter); olive, citrus, grapes, cork oak; 'orchard of the world'
Cfa / Cwa China type (warm humid)	SE USA, S China, SE Brazil, NE Argentina, E Australia	East sides at 25–40°; rain all year or in summer; mixed forests; rice, cotton, tea; typhoons/hurricanes

Köppen group · Type	Where (examples)	Climate and vegetation
Cfb Marine west coast	NW Europe (UK, France), W Canada, S Chile, NZ, Tasmania	Westerlies and warm currents (**North Atlantic Drift**) give mild winters and rain all year; deciduous forests; dairy
Temperate grasslands (steppe)	Prairies (N America), Pampas (Argentina), Steppes (Eurasia), Veld (S Africa), Downs (Australia), Canterbury (NZ)	Continental interior; dry; treeless grass; **'granaries of the world'** (wheat) and ranching
Df Taiga (boreal)	Canada, Alaska, Scandinavia, Siberia (50–70°N)	Longest and coldest winters; **coniferous forests** (pine, fir, spruce) – softwood, paper pulp, matches
ET / EF Tundra and ice cap	Arctic coasts; Greenland; **Antarctica**	Mosses, lichens, dwarf shrubs in short summer; permafrost; reindeer, Eskimo/Inuit; Antarctica is the coldest, windiest, driest continent
H Highland	Himalaya, Andes, Rockies, Tibet	Temperature and vegetation change with height (like moving towards the poles)

ANDHRA PRADESH / KURNOOL CONNECTION

India is **Am/Aw** (tropical monsoon and savanna). AP has **Aw** climate over most of the state; the **Rayalaseema/Anantapur** interior is semi-arid (BSh) because it lies in the rain-shadow of the Western Ghats. **Seshachalam–Nallamala** hills hold dry deciduous and thorn forests; **Eastern Ghats/Araku** hold moist deciduous; **Coringa and the Krishna delta** hold mangroves.

1.1 Ocean currents and climate (frequent exam link)

Warm currents	Cold currents
Gulf Stream, North Atlantic Drift, **Kuroshio** (Japan), **Brazil**, East Australian, Agulhas, Florida	**Humboldt (Peru)**, **Benguela**, **California**, **Canary**, **Labrador**, Oyashio, West Australian – cause coastal deserts (Atacama, Namib) and fog

EXAM TRAPS

Mediterranean = dry summer, wet winter (unique). **Prairies = N America**, **Pampas = Argentina**, **Veld = S Africa**, **Steppe = Eurasia**.
Hot deserts lie on the **western** sides of continents; the **Atacama** is the driest place. Wettest: **Mawsynram–Cherrapunji** (Meghalaya).

2-MINUTE REVISION

Six controls · Köppen A–E,H · Af, Aw, Am, BW, Cs, Cf, Df, ET · west coast desert, east coast humid · grasslands names · cold currents → coastal deserts.

CHECK YOURSELF

Q1. Dry summer + wet winter climate? **Mediterranean.** Q2. Veld is in? **South Africa.** Q3. Humboldt current is cold or warm? **Cold (Peru).**

CHAPTER 2 — SOCIAL GEOGRAPHY: POPULATION, SETTLEMENTS, PEOPLE

World, India and Andhra Pradesh

IN SIMPLE WORDS (what / how / why)

Social geography asks: **how many people, where, how fast are they growing, how educated, how do they live?** Population is not spread evenly: people crowd where **water, flat fertile land, mild climate and jobs** exist (river deltas, coasts) and avoid deserts, high mountains and ice. Key words: **density** = people per sq km; **sex ratio** = females per 1000 males; **literacy** = % of population aged 7+ who can read and write; **growth rate** = change per decade; **birth rate and death rate**.

2.1 Population concepts

Concept	Meaning and key points
Demographic transition	Stage 1 high birth + high death (stable); Stage 2 death falls first → rapid growth (India's recent past); Stage 3 birth falls → growth slows; Stage 4 low-low (Japan, Europe); Stage 5 ageing, decline.
Population pyramid	Wide base = young (India earlier, Africa); narrow base, bulging middle = ageing (Japan). **Dependency ratio** = (0–14 + 65+) ÷ working age (15–64). **Demographic dividend** = growth benefit when working-age share is large.
Migration	**Push** factors (poverty, war, disaster) and **pull** factors (jobs, schools); internal vs international; **rural to urban** is the biggest stream in India; marriage is the main reason for women's migration.
Human Development Index	UNDP; three dimensions: **health (life expectancy), education, income**. Top: Switzerland/Norway-type countries; India is in the 'medium' group. Also: **Gender Inequality Index, Multidimensional Poverty Index**.

2.2 India: Census 2011 key figures (learn these)

Indicator	India	Highest / lowest
Population	**121.09 crore** (17.5% of world); decadal growth **17.64%**	Most: **Uttar Pradesh** (19.98 crore), Maharashtra, Bihar · Least: **Sikkim**; UT Lakshadweep
Density (per sq km)	**382**	Highest state: **Bihar (1102)**; UT Delhi (11,297) · Lowest: **Arunachal Pradesh (17)**
Sex ratio	**943** females per 1000 males; child (0–6) **919**	Highest: **Kerala (1084)**; UT Puducherry (1037) · Lowest: **Haryana (879)**; UT Daman & Diu / Delhi low
Literacy	**74.04%** (male 82.14, female 65.46)	Highest: **Kerala (93.91%)**; Lakshadweep · Lowest: **Bihar (63.82%)**; district: Serchhip (Mizoram) highest
Urban population	**31.16%**; rural 68.84%	Most urbanised state: **Goa**; UT Delhi · Least: **Himachal Pradesh**/Bihar
SC and ST	SC **16.6%**; ST **8.6%**	ST highest %: **Mizoram, Lakshadweep, Nagaland**; SC highest %: Punjab

ANDHRA PRADESH / KURNOOL CONNECTION

Andhra Pradesh (2011, present boundaries): population about **4.95 crore**; density about **304**; sex ratio about **993** (better than India's 943); literacy about **67%** (below India's 74%); urban about **29.6%**; SC about 16.4%, ST about 5.3%; decadal growth about 9.2% (much lower than India's). Literacy is lowest in the **Vizianagaram–Srikakulam tribal belt**. Next census: the 2027 census with caste enumeration has been announced; its results are not yet available.

2.3 Settlements, urbanisation, tribes, language, religion

Topic	Key facts
Rural settlements	**Compact/nucleated** (Ganga plain, deltas – water and defence), **semi-compact**, **dispersed/isolated** (hills, forests, Meghalaya, Kerala), **linear** (along roads/rivers/canals).
Urban terms	**Town**: ≥5000 people, density ≥400/sq km, 75% male workers non-farm. **Million-plus city**; **Urban agglomeration** = city + outgrowths. Largest UAs: **Greater Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad**. Schemes: **Smart Cities (2015), AMRUT, HRIDAY**.
Tribes	Largest STs: **Gonds, Bhils, Santhals**. **PVTGs** (particularly vulnerable tribal groups) – 75 in India. **AP tribes: Chenchu** (Nallamala; hunter-gatherers; PVTG), **Savara, Khond (Kondh), Gadaba, Jatapu, Konda Reddy, Yanadi, Yerukala, Sugali/Lambada, Koya, Valmiki**. Tribal areas: **Araku, Paderu, Rampachodavaram, Seethampeta**. ITDAs run tribal development.
Language	**22 scheduled languages** (8th Schedule); Hindi most spoken (about 44% mother tongue); **Telugu is the 4th most spoken** (after Hindi, Bengali, Marathi) and the largest Dravidian language. **Classical languages:** Tamil (2004), Sanskrit (2005), **Telugu and Kannada (2008)**, Malayalam (2013), Odia (2014), and in 2024 Marathi, Pali, Prakrit, Assamese, Bengali.
Religion (2011)	Hindus 79.8%, Muslims 14.2%, Christians 2.3%, Sikhs 1.7%, Buddhists 0.7%, Jains 0.4%.

EXAM TRAPS

Sex ratio is **females per 1000 males** (not %). **Density ≠ population** – Bihar is densest state but UP has most people.
Kerala tops literacy and sex ratio; Haryana is lowest in sex ratio; Bihar lowest in literacy; Arunachal lowest density; Sikkim least populous state.

2-MINUTE REVISION

India 121.09 cr · density 382 · SR 943 · literacy 74.04 · urban 31.16 · SC 16.6 · ST 8.6 · Kerala/Bihar/Haryana/UP/Sikkim/Arunachal extremes · AP: SR 993, literacy ~67 · Chenchu PVTG · Telugu classical 2008.

CHECK YOURSELF

Q1. Which state has the highest sex ratio? **Kerala.** Q2. PVTG of the Nallamala forest? **Chenchu.** Q3. Telugu got classical status in? **2008.**

CHAPTER 3 — ECONOMIC GEOGRAPHY I: AGRICULTURE

World types, Indian seasons and crops, revolutions

IN SIMPLE WORDS (what / how / why)

Farming is the biggest user of land and people in India (about 45% of workers). A crop grows where three things meet: **right temperature, enough water and suitable soil**. Hence rice needs hot-humid delta soils, wheat needs cool winter and moderate rain, cotton needs black soil and a long warm season, tea needs hill slopes with no waterlogging.

Type of farming (world)	Features and examples
Shifting cultivation	Slash and burn; **Jhum** (NE India), **Ladang** (Malaysia), **Milpa** (Central America), **Conuco** (Venezuela).
Intensive subsistence	Monsoon Asia: small plots, high labour, rice; India, China, Bangladesh.
Commercial grain	Large mechanised farms: prairies, steppes, pampas, Canterbury – wheat and maize.